

Foreign Giants' Broad Shoulders

The Causal Effect of Foreign Ideas on US Innovation

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Introduction

Motivation

- Core ideas in economic growth theory

1. Ideas build on ideas (Schumpeter, 1911; Romer, 1990; Aghion-Howitt, 1992)
2. Ideas diffuse across space (Griliches, 1957; Jaffe et al., 1993)

- Missing in the data:

1. The **production structure of ideas**: which ideas are inputs into which new ideas?
2. The **geography of knowledge**: which ideas are available in a given location?

This paper:

- A. Constructs a directed **genealogy of technologies** (an input–output network of ideas) from the full text of USPTO patents 1890-2019
- B. Measures local access to **foreign knowledge** at the technology level
- C. Quantifies how access to foreign ideas shapes U.S. innovation

Main Findings

1. **U.S. innovation depends heavily on foreign ideas**

- ▶ 67% of recent U.S. technologies rely on at least one foreign precursor
- ▶ Dependence is significantly stronger for **breakthrough** technologies

2. **Common ancestry shapes access to foreign ideas**

- ▶ Counties with more ancestry from a country are more likely to absorb its technologies

3. **Foreign ideas have large causal effects on U.S. innovation and jobs**

- ▶ Doubling ancestry $\Rightarrow$ +21% absorption of foreign technologies
- ▶ Doubling absorption $\Rightarrow$ +41% downstream innovation, +280% job creation

4. **Local idea stocks determine the direction of innovation.**

- ▶ Places are more likely to invent technologies whose inputs are already locally available

Access to foreign knowledge shapes both where and what gets invented in the U.S.

Contribution

- Ideas build on ideas (Schumpeterian growth)

Schumpeter (1911), Romer (1990), Aghion & Howitt (1992), Weitzman (1998), Jones (2009)

Provide a direct empirical measure of the production structure of ideas.

- Measures of technological proximity

Jaffe (1986), Hall, Jaffe & Trajtenberg (2001), Kelly et al. (2021)

Move from proximity in patent space to directed dependence across technologies.

- Immigration stimulates innovation

Kerr (2008), Hunt & Gauthier-Loiselle (2010), Moser & al. (2014), Posch & al. (2025), Terry et al. (2025)

Co-ethnicity shapes the direction of innovation.

- Effects of local innovation on local economies

van Reenen (1997), Moretti (2010), Kline & al. (2019), Kalyani & al. (2025)

Causal effect of foreign idea absorption on local innovation and employment.

Foreign Technologies

Technologies Data (Kalyani et al., 2025, QJE)

- Minimally adjust text-based methodology to extend back in time and space

1. Identify technologies using Wikipedia and US patents, 1834–2023

- ▶ Extract n-grams from the titles and first 50 words of summaries of Wikipedia pages
 - Remove n-grams with stop words, and those found in COHA before 1830
- ▶ Match each ngram to a Wikipedia page and keep it if that page describes a technology
- ▶ Group matched bigrams into “technologies” (Wikipedia titles)

2. Assign origin country: first country to reach 10 patents mentioning the technology

3. Define birth year $b(\tau)$: first year this threshold is reached in any country

4. Flag mentions of each new technology in $\sim 250\text{M}$ *Lightcast* job postings 2010–2023.

Foreign and US Technologies 1900–2019

- Identify 19,542 technologies
- 1,386 (7.1%) originate abroad
- Propensity to patent in US varies by country → address econometrically

Country	Technologies	Share
Japan	567	40.91
Germany	310	22.37
United Kingdom	157	11.33
France	59	4.26
Canada	53	3.82
Switzerland	45	3.25
South Korea	30	2.16
Others	165	11.90
Total	1,386	100.00

Examples of Foreign Technologies by Country

Country	Bigram	Title	Birth Year
Japan	cd rom	CD-ROM	1986
Japan	hybrid electric vehicle	Hybrid Car / Hybrid Electric Vehicle	1991
Germany	electron microscope	Electron Microscope	1939
Germany	polyvinyl chloride	Polyvinyl Chloride (PVC)	1936
Netherlands	wireless lan	Wireless LAN	1992
United Kingdom	specific antigen	Antigen-Antibody Interaction	1970
Sweden	cloud computing	Cloud Computing	2008
South Korea	blu-ray ripper	Blu-ray	2002
Finland	service gpr	GPRS Roaming Exchange	1997

A Genealogy of Technologies

Constructing the Genealogy of Technologies (1/2)

Key advantage

- Observe **technologies** directly (rather than relying on citations or similarity measures)
⇒ Can track which technologies appear in the **same patents**

- For any technology τ , define all patents mentioning it

$$\mathcal{P}(\tau) = \{\text{patents mentioning } \tau\}$$

- For each technology τ' , define **founding patents**

$$\mathcal{F}(\tau') = \{\text{patents mentioning } \tau' \text{ within 5 years of birth}\}$$

Constructing the Genealogy of Technologies (2/2)

Construction:

- Define dependence

$$s_{\tau \rightarrow \tau'} = \frac{|\mathcal{F}(\tau') \cap \mathcal{P}(\tau)|}{|\mathcal{F}(\tau')|}$$

- Define link

$$\begin{aligned} \tau \rightarrow \tau' \quad & \text{if } s_{\tau \rightarrow \tau'} \geq 0.10 \\ & \text{and } b(\tau) < b(\tau') \end{aligned}$$

Interpretation:

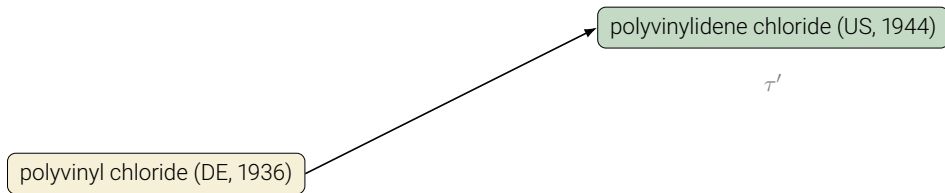
- τ is a **parent** (precursor) of τ'
- Directed link = **input into production of new ideas**
- Yields a large-scale $19,542 \times 19,542$ **directed network** describing technological development in the U.S. since 1900

Example Foreign Parent Technology

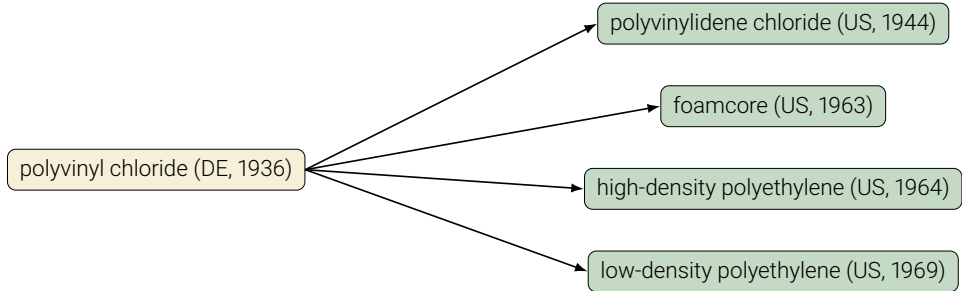
polyvinyl chloride (DE, 1936)

τ

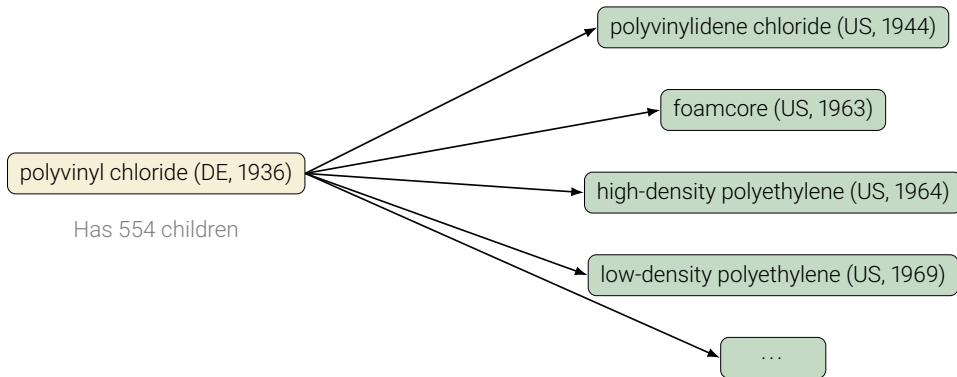
Example Foreign Parent Technology



Example Foreign Parent Technology

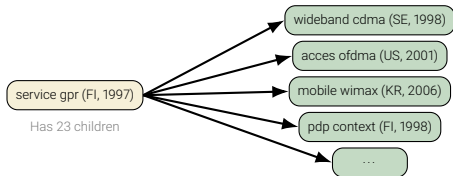
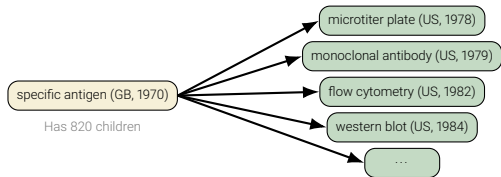
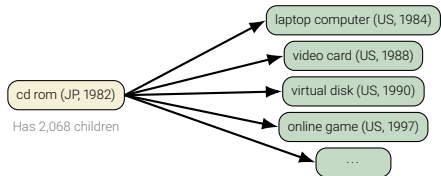
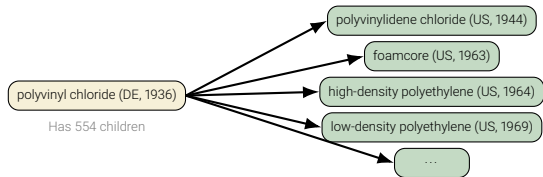


Example Foreign Parent Technology

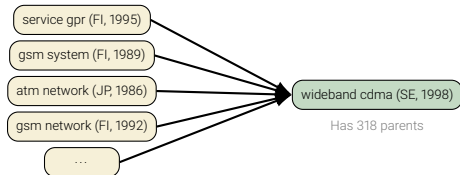
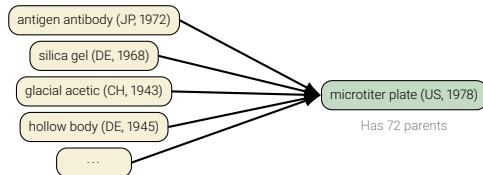
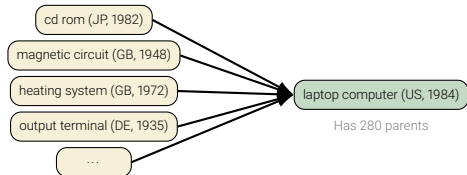
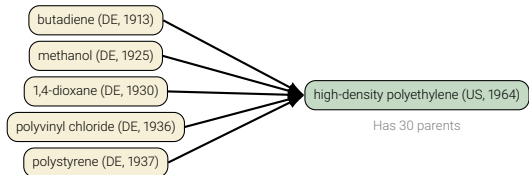


- ▶ The average US technology has 50 parents, 3 of which are foreign
- ▶ 67% of recent US technologies have at least one foreign parent

Example Foreign Parent Technologies

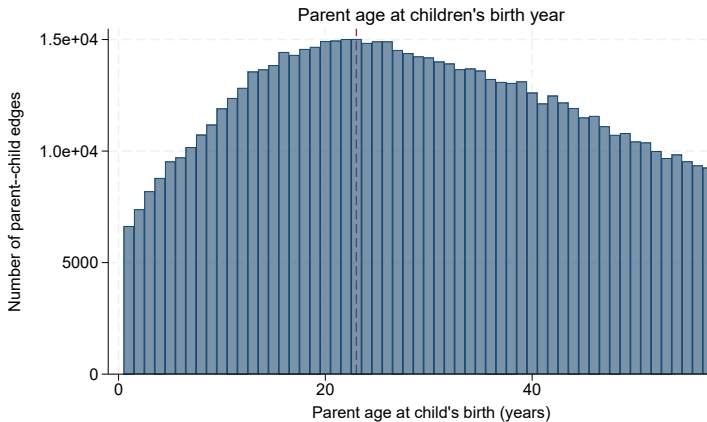


Example Technologies with Many Foreign Parents



Parent Age at Children's Birth Year

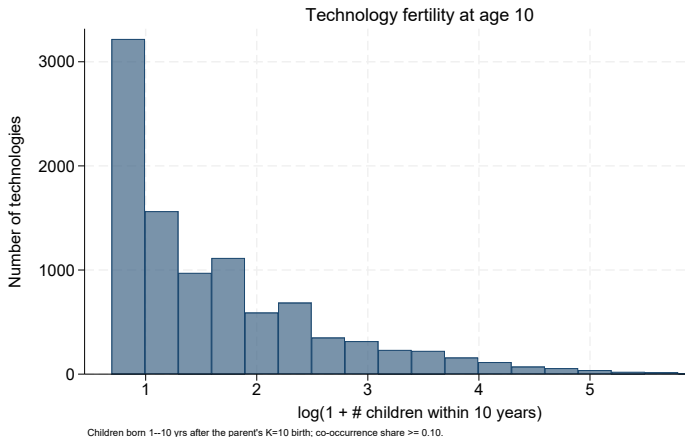
- Technologies exhibit a life-cycle
- Fertility is spread over decades, peaking about 23 years after birth



Genealogy edges, co-occurrence share ≥ 0.10 ; birth = K=10 year. Modal age = 23 years.

Technology Fertility at Age 10

- Fertility is **highly skewed**
- Median: 3 children
- Mean: 10 children
- 99th percentile: 126 children
- Define **breakthrough technologies** as those with a high ratio (B) of descendants to number of parents



Breakthrough Technologies by Birth Year, $B \geq 25$ (1/2)

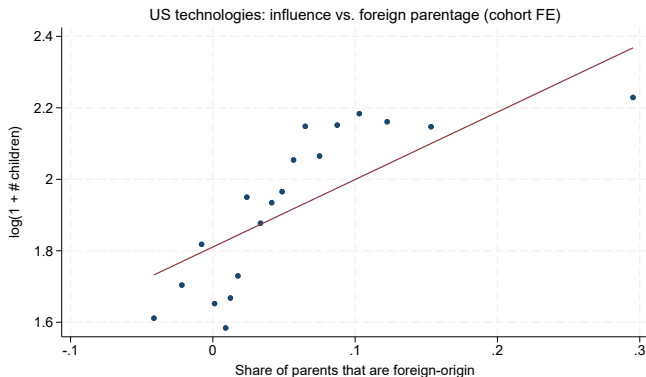
Year	Bigram	Title	Country	B	Year	Bigram	Title	Country	B
1933	reaction condition	chemical reaction	DE	32.6	1958	laser beam	laser cutting	US	44.5
1933	voltage source	power electronics	DE	33.4	1959	layer chromatogr..	thin-layer chromatogr..	US	31.2
1933	metal surface	chemical coloring of m..	GB	42.0	1960	emitting diode	laser diode	US	50.4
1934	synthetic rubber	synthetic rubber	US	41.7	1965	signal input	phase detector	US	148.0
1935	ground potential	residual-current device	DE	32.7	1968	memory ram	static random-access m..	US	56.1
1936	time delay	time delay and integr..	US	69.8	1971	floppy disk	superdisk	US	75.0
1939	input terminal	operational amplifier	DE	45.0	1973	specific binding	elisa	US	31.0
1941	annular space	production packer	DE	43.5	1974	recombinant dna	molecular cloning	US	35.0
1944	printed circuit	printed circuit board	US	29.5	1974	dna fragment	southern blot	DE	31.6
1946	semiconductor d..	semicond. device fab..	US	29.8	1975	endothelial cell	angiogenesis	US	58.5
1948	data processing	data processing	US	59.5	1975	monoclonal antibody	monoclonal antibody	US	30.9
1948	epoxy resin	epoxy	US	73.5	1975	dna sequence	dna sequencing	FR	33.3
1949	earth metal	alkaline earth metal	GB	56.2	1977	personal computer	personal computer	US	259.5
1951	insulating mater..	thermal insulation	DE	263.0	1977	hard disk	hard disk recorder	US	111.7
1956	integrated circuit	die (integrated circuit)	US	140.8	1977	protease inhibitor	protease	JP	40.0

Breakthrough Technologies by Birth Year, $B \geq 25$ (2/2)

Year	Bigram	Title	Country	B	Year	Bigram	Title	Country	B
1978	touch screen	multi-touch	US	31.8	1986	client server	client (computing)	US	29.1
1978	user interface	user interface	US	68.5	1987	specific activity	specific activity	US	48.7
1978	word processor	pages (word processor)	JP	37.3	1987	transmitted signal	continuous-wave radar	GB	32.0
1979	cultured cell	sulforhodamine b	JP	49.5	1988	markup language	xml	US	26.9
1980	network lan	wireless lan	US	110.8	1988	transfer protocol	file transfer protocols	US	52.8
1981	service provider	service provider	US	102.0	1989	internet protocol	internet control msg..	US	39.8
1981	database system	database	US	45.0	1990	personal digital	personal digital assist..	US	43.8
1982	cd rom	super nes cd-rom	JP	207.3	1991	plug in	openmediavault	US	269.0
1982	cellular telephone	plain old telephone s..	US	34.5	1992	control element	surf (web browser)	DE	45.3
1984	graphical user	user interface	US	67.9	1994	learning algorithm	machine learning	US	
1984	object oriented	OO programming	US	81.0	1995	smart phone	smartphone	US	
1984	relational database	Amazon RDS	US	39.8	1997	drug eluting	drug-eluting stent	US	
1984	specific integra..	semiconductor IP core	US	82.3	2000	blu ray	blu-ray	JP	
1985	flash memory	compactflash	US	111.2	2003	social medium	social media	US	
1985	desktop computer	desktop computer	US	100.5	2004	cloud computing	cloud computing	SE	

Foreign Parentage and Technological Influence

- US technologies with a higher share of **foreign-origin parents** go on to produce more child technologies
 - Binscatter of $\ln(1 + \# \text{ children})$ on the foreign-origin share of a technology's own parents
- Holds **within birth-decade cohorts**:
slope 1.89 (SE 0.14), $p < 0.001$;
 $N = 17,259$ US technologies



Ancestry and the Local Absorption of Foreign Technologies

From Genealogy to Geography

Genealogy shows:

- Technologies build on each other
- Foreign technologies are key inputs into U.S. innovation

Are some U.S. locations better at absorbing foreign ideas?

- Existing work shows co-ethnicity transmits origin-specific knowledge and facilitates cross-border economic linkages
- $\Rightarrow$ Ancestry from country of origin o is a natural proxy for local access to ideas from o

Local Absorption of Foreign Technologies

Track where foreign technologies are mentioned across U.S. locations

For each technology τ , origin country o , U.S. county d , and year t :

- $P_{\tau,o,d,t}$ Number of patents in d at t that mention technology τ from origin o

Interpretation:

- $P_{\tau,o,d,t} > 0 \Rightarrow$ technology τ from o is absorbed in d at t
- $P_{o,d,t} \equiv \sum_{\tau \in o} P_{\tau,o,d,t}$: local absorption of technologies from o in county d

Hypothesis:

- Counties with more ancestry from o absorb more technologies from o .
- Measure $A_{o,d}$ from the US Census

Exogenous Variation in Ancestry

$$A_{o,d} = \delta_o + \delta_d + \sum_{\tau=1880}^t \beta^\tau \underbrace{I_o^\tau}_{\text{push}} \times \underbrace{\frac{I_d^\tau}{I^\tau}}_{\text{pull}} + \nu_{o,d}$$

[Buchardi et al. 18';

Bursztyn et al. 24'; Terry et al. 25']

■ Push-Pull Instrument:

- ▶ **Push** All migrants arriving in U.S. from origin o at time τ
- ▶ **Pull** Fraction immigrants from other origins settling in destination county d at time t

■ The stock of ancestry accumulates with immigration flows

⇒ Instrument $A_{o,d}$ with historical interactions of push and pull factors

■ Use broad leave-out categories:

- ▶ **Push** All migrants leaving o but settling in regions other than $r(d)$
- ▶ **Pull** Fraction immigrants leaving continents other than $c(o)$ settling in d

Estimation Strategy

Second Stage

$$\ln(1 + P_{o,d}) = \delta_o + \delta_d + \delta_o \times \delta_{r(d)} + \beta \ln(\widehat{1 + A_{o,d}}) + X'_{o,d} \zeta + \varepsilon_{o,d}$$

- $P_{o,d}$ are US inventors' patents mentioning technologies from country o in county d

First Stage

$$\ln(1 + A_{o,d}) = \delta_o + \delta_d + \delta_o \times \delta_{r(d)} + \sum_{t=1880}^T \beta^t I_{o,-r(d)}^t \frac{I_{-c(o),d}^t}{I_{-c(o)}^t} + X'_{o,d} \eta + \nu_{o,d}$$

- $\delta_o \times \delta_{r(d)}$ are origin-region fixed effects
- $X_{o,d}$ are log distance and latitude differences polynomials controls

Causal Effect of Co-Ethnicity on Idea Diffusion

	(1)	(2)	(3)	(4)
	Log # Foreign Techs	Log # Foreign Techs	Log # Foreign Techs	Log US Patents Mentioning Foreign Techs
Log Ancestry _{o, d}	0.185*** (0.012)	0.245*** (0.018)	0.299*** (0.022)	0.730*** (0.052)
Observations	606213	606213	606213	606213
Estimates	OLS	IV	IV	IV
Origin countries	All 193	All 193	All 193	All 193
Clustering	State	State	State	State
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Origin-Region FE	No	No	Yes	Yes
Distance control	No	No	Yes	No
KP first-stage F	.	261.060	107.287	106.956
Stock-Yogo (10% size)	.	43.270	43.270	43.270
Anderson-Rubin F		118.582	108.919	120.166
Anderson-Rubin p		0.000	0.000	0.000

► First Stage

► Robustness (1/2)

► Robustness (2/2)

- Common ancestry is a major driver of domestic absorption of foreign ideas

The Diffusion of Foreign Technologies and US Innovation

The Causal Effect of Foreign Ideas on US Innovation

	Log # US Child Techs from Origin o , Invented 1980–1990		
	(1)	(2)	(3)
Log Ancestry $_{o,d}$	0.157*** (0.022)	-0.030** (0.014)	
Log Non-Generative Absorption 1980 (technologies) $_{o,d}$		0.643*** (0.106)	0.588*** (0.114)
Observations	606213	606213	606213
Outcome window	1980–1990	1980–1990	1980–1990
Non-zero outcome cells	722.000	722.000	722.000
Clustering	State	State	State
Clusters	51.000	51.000	51.000
Origin FE	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes
Distance Control	Yes	Yes	Yes
Kleibergen–Paap F	259.591	43.539	47.339
SW F (Ancestry)		283.713	
SW F (Non-Gen)		94.979	
Anderson–Rubin F	120.480	120.480	120.480
Anderson–Rubin p	0.000	0.000	0.000

► First Stage

- Local absorption of foreign ideas causes local invention of child technologies
- Ethnic composition of US counties is key to their ability to innovate

	Log # US Child Techs from Origin o , Invented 1980–1990			
	(1)	(2)	(3)	(4)
Log Non-Generative Absorption 1980– o , d	0.588*** (0.114)	0.559*** (0.112)	0.465*** (0.121)	0.549*** (0.117)
Log Foreign-born 2010– o , d		0.054*** (0.007)		
Log Foreign-born 1970– o , d			0.142*** (0.033)	
FDI Dummy– o , d				0.096*** (0.034)
Observations	606213	606213	606213	606213
Standard errors	State	State	State	State
Outcome window	1980–1990	1980–1990	1980–1990	1980–1990
Clusters	51.000	51.000	51.000	51.000
Kleibergen–Paap F	47.339	75.237	69.539	45.679
SW F (Non-Gen)		79.338	223.855	68.645
SW F (Second endog.)		64.054	689.894	56.861
Anderson–Rubin F	120.480	120.480	120.480	120.480
Anderson–Rubin p	0.000	0.000	0.000	0.000
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Distance Control	Yes	Yes	Yes	Yes

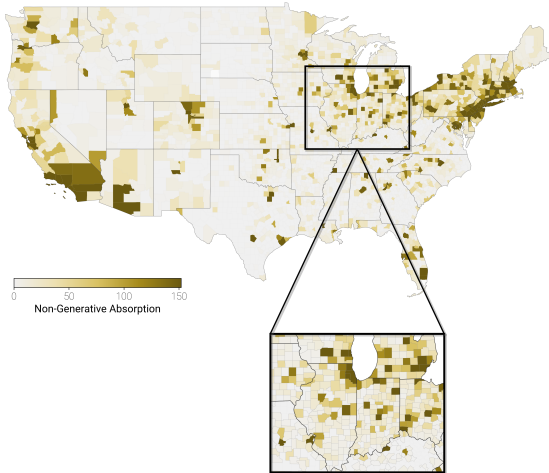
► First Stage

- Effect of co-ethnicity transmits through the local absorption of foreign technologies

Germany

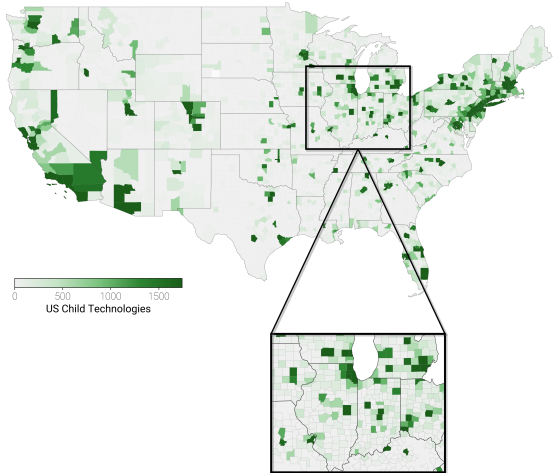
Non-Generative Absorption

Polyvinyl Chloride, Electron Microscope, Gas Turbine, Frequency Modulation



US Child Technologies

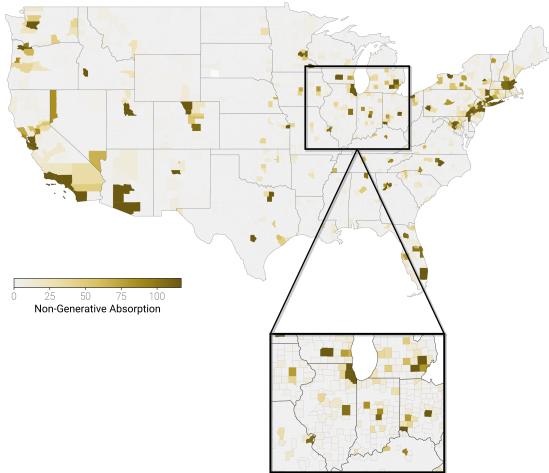
Plastic Film, Density Polyethylene, Carbon Fiber, Medical Device



Japan

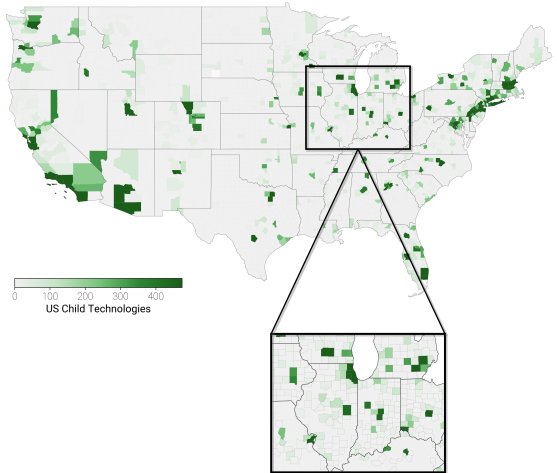
Non-Generative Absorption

CD-ROM, Hybrid Car, Compact Disc, Blu-Ray



US Child Technologies

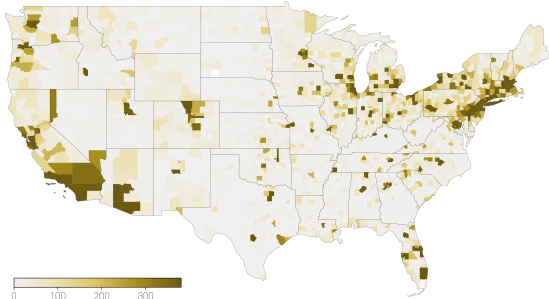
Laptop Computer, Video Card, Virtual Disk, Online Game



All Countries

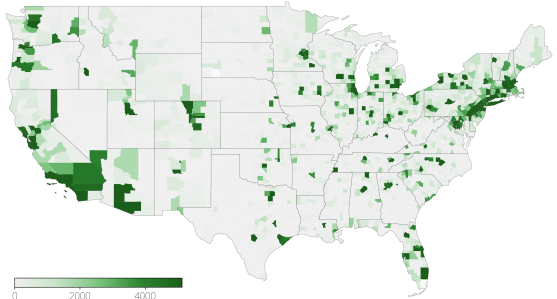
Non-Generative Absorption

CD-ROM, Polyvinyl Chloride, Specific Antigen, Service GPR



US Child Technologies

Laptop Computer, Medical Device, Monoclonal Antibody, Wideband CDMA



The Diffusion of Foreign Technologies and US Jobs

Effect of Foreign Idea Diffusion on US Jobs

	Log Job Postings from Origin o in County d		
	(1)	(2)	(3)
Log Non-Generative Absorption 1980 $_{o,d}$	2.665*** (0.184)	4.102*** (0.419)	4.036*** (0.406)
Observations	606213	606213	606213
Outcome	Foreign-tech postings	Child postings, all	Child postings, born 1980+
Standard errors	State	State	State
Clusters	51.000	51.000	51.000
Kleibergen–Paap F	47.339	47.339	47.339
Anderson–Rubin F	20.860	22.294	22.220
Anderson–Rubin p	0.000	0.000	0.000
Origin FE	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes
Distance Control	Yes	Yes	Yes

► First Stage

US counties that absorb foreign technologies have

- More jobs using imported technologies
- More jobs in using US innovations that build on those imported technologies

Local Idea Stocks and the Direction of Technical Change

Predicting the Direction Innovation from Local Idea Stock

- Can local absorption predict where in the US **specific** new technologies are invented?
- For each US technology τ' and county d , measure

Inputs Available $_{\tau',d}$ = parent technologies locally absorbed before τ' was born

- Are counties with greater local access to those inputs more likely to invent τ' ?

$$\mathbb{1}[P_{\tau',d} > 0] = \alpha + \beta \text{ \# Inputs Available}_{\tau',d} + \varepsilon_{\tau',d}$$

- $P_{\tau',d}$ = patents mentioning τ' in county d within 10 years of birth
- β measures path dependence at the technology level.

Direction of Local Innovation

	Local Child Invention			
	(1)	(2)	(3)	(4)
	LPM	LPM	LPM	LPM-IV
Log(1+# inputs available)	0.0053*** (0.0002)			
Log(1+# US inputs available)		0.0035*** (0.0002)		
Log(1+# foreign inputs available)		0.0066*** (0.0004)		
MI available			0.0108*** (0.0003)	0.1371*** (0.0056)
Observations	85993371	85993371	85993371	85993371
KP first-stage F				868.3601

Direction of Local Innovation – Robustness to Parent Designation

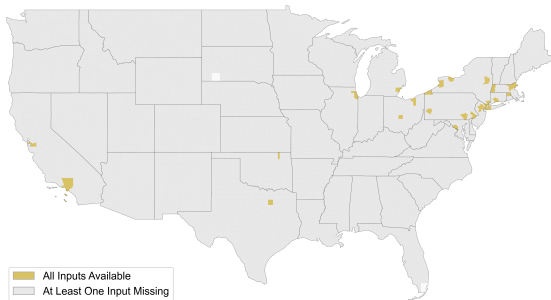
	Local Child Invention			
	(1)	(2)	(3)	(4)
	Most Infl.	Most Recent	Single ctry.	Modal
Input available	0.1371*** (0.0056)	0.1858*** (0.0068)	0.1280*** (0.0071)	0.0771*** (0.0071)
Observations	85993371	85993371	41990559	85993371
Designation	Most Influential	Most Recent	Single country	Modal
Origin FE	Yes	Yes	Yes	Yes
County FE	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes
Distance Polynomials	No	No	No	No
KP first-stage F	868.3601	624.5181	495.1474	1.1e+03

- The effect is robust to how each technology's foreign parent is designated – most influential, most recent, single-origin, or modal country

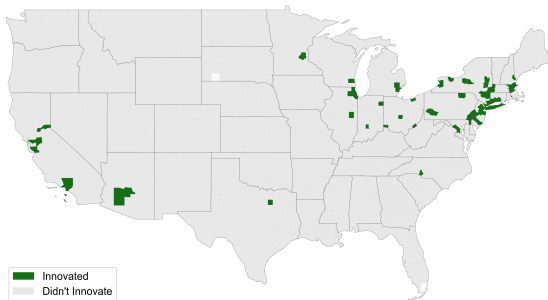
Semiconductor Material (1948)

All Recently Invented Foreign Inputs Available (1938–1948):

Conductive Material (DE), Insulating Material (DE), Input Terminal (DE),
Output Terminal (DE), Periodic Table (DE), Voltage Source (DE)



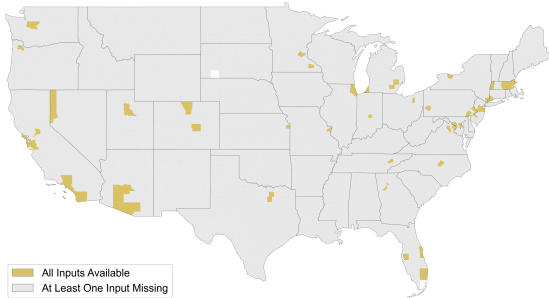
Founding Patents of Semiconductor Material (1948–1958)



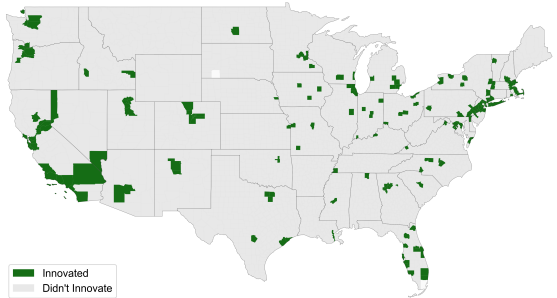
Cell Phone (1995)

All Recently Invented Foreign Inputs Available (1985–1995):

Communication System (DE), Operating System (GB), Radio Frequency (GB), CD ROM (JP)



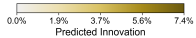
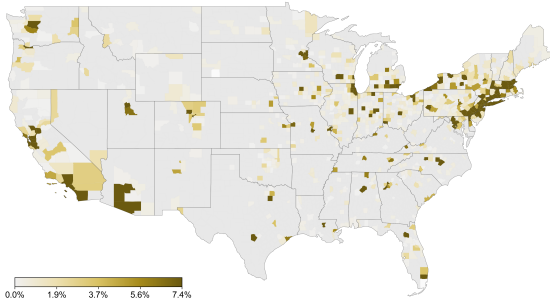
Founding Patents of Cell Phone (1995–2005)



Top 500 Breakthrough Technologies

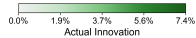
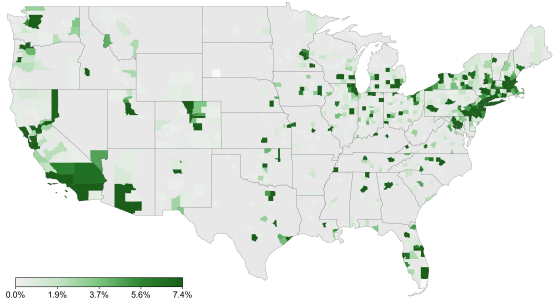
Predicted Innovation Probability

Control Function Logit



Actual Innovation Rate

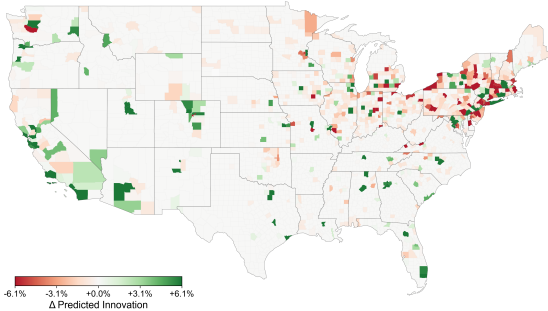
Control Function Logit



Winners and Losers Across Eras (1980 Split)

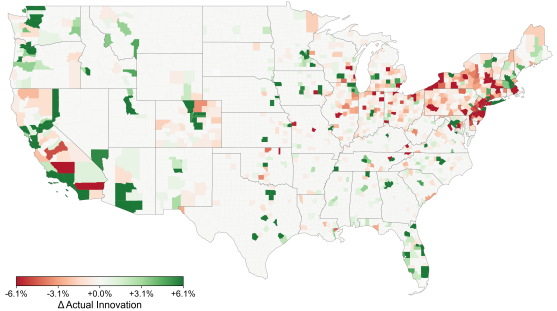
Change in Predicted Innovation

Control Function Logit



Change in Actual Innovation

Control Function Logit



- Change in origin countries of foreign ideas aligns with a marked shift of innovative activity away from the heartlands and towards the coasts post-1980

Conclusion

Conclusion

■ **Measure the production structure of ideas**

- ▶ Construct a directed genealogy of technologies from patent text
- ▶ Identify which new technologies build which specific prior ideas

■ **U.S. innovation is fundamentally global**

- ▶ Most U.S. technologies rely on foreign inputs
- ▶ Foreign ideas are key inputs into breakthrough innovations

■ **Local access to foreign ideas generates a comparative advantage in innovation**

- ▶ Ancestry determines local access to origin-specific knowledge
- ▶ Foreign ideas causally drive innovation and job creation

■ **Local idea stocks determine the direction of technological change**

- ▶ Places invent the technologies their existing ideas allow them to recombine

Appendix

	Log Ancestry			
	(1)	(2)	(3)	(4)
$I_{o,-r(d)}^{1880} \times \frac{I_{-c(o),d}^{1880}}{I_{-c(o)}^{1880}}$	0.038*** (0.013)	0.023** (0.011)	0.023** (0.011)	0.040*** (0.010)
$I_{o,-r(d)}^{1900} \times \frac{I_{-c(o),d}^{1900}}{I_{-c(o)}^{1900}}$	0.021 (0.046)	0.037 (0.032)	0.039 (0.032)	0.090*** (0.034)
$I_{o,-r(d)}^{1910} \times \frac{I_{-c(o),d}^{1910}}{I_{-c(o)}^{1910}}$	0.117*** (0.039)	0.096*** (0.029)	0.095*** (0.029)	0.090* (0.048)
$I_{o,-r(d)}^{1920} \times \frac{I_{-c(o),d}^{1920}}{I_{-c(o)}^{1920}}$	0.151*** (0.056)	0.164*** (0.045)	0.160*** (0.044)	0.217*** (0.047)
$I_{o,-r(d)}^{1930} \times \frac{I_{-c(o),d}^{1930}}{I_{-c(o)}^{1930}}$	-0.194** (0.081)	-0.205*** (0.065)	-0.202*** (0.064)	-0.152*** (0.048)
$I_{o,-r(d)}^{1970} \times \frac{I_{-c(o),d}^{1970}}{I_{-c(o)}^{1970}}$	0.144*** (0.034)	0.132*** (0.030)	0.131*** (0.030)	0.112*** (0.024)
$I_{o,-r(d)}^{1980} \times \frac{I_{-c(o),d}^{1980}}{I_{-c(o)}^{1980}}$	0.067** (0.027)	0.073*** (0.026)	0.074*** (0.026)	0.120*** (0.020)
Kleibergen-Paap Wald RK F Stat	20.461	25.239	24.916	167.513
Stock-Yogo 5% Critical Values	19.86	19.86	19.86	21.01
Stock-Yogo 10% Critical Values	11.29	11.29	11.29	11.52
Observations	606,213	606,213	606,213	606,213
Origin FE	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes
Origin-Region FE	No	Yes	Yes	Yes
Distance Polynomials	No	No	Yes	Yes
Principal Components	No	No	No	Yes

Standard errors clustered at the country level.

	(1)	(2)	(3)
	Ancestry	Ancestry	Non-Gen
$I_{a,-r(d)}^{1880} \times \frac{I_{-c(o),d}^{1880}}{I_{-c(o)}^{1880}}$	0.040*** (0.010)	0.040*** (0.010)	0.001 (0.025)
$I_{a,-r(d)}^{1900} \times \frac{I_{-c(o),d}^{1900}}{I_{-c(o)}^{1900}}$	0.090*** (0.034)	0.090*** (0.034)	0.101* (0.057)
$I_{a,-r(d)}^{1910} \times \frac{I_{-c(o),d}^{1910}}{I_{-c(o)}^{1910}}$	0.090* (0.048)	0.090* (0.048)	0.021 (0.072)
$I_{a,-r(d)}^{1920} \times \frac{I_{-c(o),d}^{1920}}{I_{-c(o)}^{1920}}$	0.217*** (0.047)	0.217*** (0.047)	0.015 (0.064)
$I_{a,-r(d)}^{1930} \times \frac{I_{-c(o),d}^{1930}}{I_{-c(o)}^{1930}}$	-0.152*** (0.048)	-0.152*** (0.048)	-0.018 (0.088)
$I_{a,-r(d)}^{1970} \times \frac{I_{-c(o),d}^{1970}}{I_{-c(o)}^{1970}}$	0.112*** (0.024)	0.112*** (0.024)	0.188*** (0.045)
$I_{a,-r(d)}^{1980} \times \frac{I_{-c(o),d}^{1980}}{I_{-c(o)}^{1980}}$	0.120*** (0.020)	0.120*** (0.020)	-0.057** (0.025)
Kleibergen-Paap Wald RK F Stat	167.513	20.837	12.400
Stock-Yogo 5% Critical Values	21.01	19.40	21.01
Stock-Yogo 10% Critical Values	11.52	10.78	11.52
Observations	606,213	606,213	606,213
Origin FE	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes
Origin-Region FE	Yes	Yes	Yes
Distance Polynomials	Yes	Yes	Yes
Principal Components	Yes	Yes	Yes

Standard errors clustered at the country level.

	(1)	(2)		(3)		(4)		(5)	
	Non-Gen	Non-Gen	FB 2010	Non-Gen	FB 1970	Non-Gen	FDI	Non-Gen	Info Dem.
$I_{a,-r(d)}^{1880} \times \frac{I_{-c(a),d}^{1880}}{I_{-c(a)}^{1880}}$	0.001 (0.025)	0.001 (0.025)	0.006** (0.003)	0.001 (0.025)	0.010 (0.006)	0.001 (0.025)	0.008** (0.004)	0.076* (0.044)	0.038 (0.025)
$I_{a,-r(d)}^{1900} \times \frac{I_{-c(a),d}^{1900}}{I_{-c(a)}^{1900}}$	0.101* (0.057)	0.101* (0.057)	0.012 (0.011)	0.101* (0.057)	0.044*** (0.015)	0.101* (0.057)	0.031*** (0.011)	0.013 (0.101)	0.068 (0.075)
$I_{a,-r(d)}^{1910} \times \frac{I_{-c(a),d}^{1910}}{I_{-c(a)}^{1910}}$	0.021 (0.072)	0.021 (0.072)	0.016 (0.019)	0.021 (0.072)	0.048*** (0.017)	0.021 (0.072)	0.032*** (0.010)	0.082 (0.088)	0.210* (0.122)
$I_{a,-r(d)}^{1920} \times \frac{I_{-c(a),d}^{1920}}{I_{-c(a)}^{1920}}$	0.015 (0.064)	0.015 (0.064)	0.053*** (0.017)	0.015 (0.064)	0.053** (0.024)	0.015 (0.064)	0.050** (0.020)	-0.046 (0.048)	0.135*** (0.051)
$I_{a,-r(d)}^{1930} \times \frac{I_{-c(a),d}^{1930}}{I_{-c(a)}^{1930}}$	-0.018 (0.088)	-0.018 (0.088)	-0.037 (0.025)	-0.018 (0.088)	-0.012 (0.031)	-0.018 (0.088)	-0.047** (0.023)	0.256* (0.134)	-0.177* (0.106)
$I_{a,-r(d)}^{1970} \times \frac{I_{-c(a),d}^{1970}}{I_{-c(a)}^{1970}}$	0.188*** (0.045)	0.188*** (0.045)	0.032** (0.013)	0.188*** (0.045)	0.083*** (0.021)	0.188*** (0.045)	0.024** (0.010)	0.069 (0.048)	0.057** (0.028)
$I_{a,-r(d)}^{1980} \times \frac{I_{-c(a),d}^{1980}}{I_{-c(a)}^{1980}}$	-0.057** (0.025)	-0.057** (0.025)	0.210*** (0.035)	-0.057** (0.025)	0.054*** (0.011)	-0.057** (0.025)	0.030*** (0.006)	-0.044 (0.027)	0.039 (0.054)
Kleibergen-Paap Wald RK F Stat	11.400	13.681		23.967		285.368		34.647	
Stock-Yogo 5% Critical Values	21.01	19.40		19.40		19.40		19.83	
Stock-Yogo 10% Critical Values	11.52	10.78		10.78		10.78		10.89	
Observations	606,213	606,213		606,213		606,213		19,110	
Origin FE	Yes	Yes		Yes		Yes		Yes	
Destination FE	Yes	Yes		Yes		Yes		Yes	
Principal Components	Yes	Yes		Yes		Yes		Yes	

Standard errors clustered at the country level.

	(1)	(2)
	Non-Gen	Non-Gen
$I_{a,-r(d)}^{1880} \times \frac{I_{-c(a),d}^{1880}}{I_{-c(a)}^{1880}}$	0.001 (0.025)	0.001 (0.025)
$I_{a,-r(d)}^{1900} \times \frac{I_{-c(a),d}^{1900}}{I_{-c(a)}^{1900}}$	0.101* (0.057)	0.101* (0.057)
$I_{a,-r(d)}^{1910} \times \frac{I_{-c(a),d}^{1910}}{I_{-c(a)}^{1910}}$	0.021 (0.072)	0.021 (0.072)
$I_{a,-r(d)}^{1920} \times \frac{I_{-c(a),d}^{1920}}{I_{-c(a)}^{1920}}$	0.015 (0.064)	0.015 (0.064)
$I_{a,-r(d)}^{1930} \times \frac{I_{-c(a),d}^{1930}}{I_{-c(a)}^{1930}}$	-0.018 (0.088)	-0.018 (0.088)
$I_{a,-r(d)}^{1970} \times \frac{I_{-c(a),d}^{1970}}{I_{-c(a)}^{1970}}$	0.188*** (0.045)	0.188*** (0.045)
$I_{a,-r(d)}^{1980} \times \frac{I_{-c(a),d}^{1980}}{I_{-c(a)}^{1980}}$	-0.057** (0.025)	-0.057** (0.025)
Kleibergen-Paap Wald RK F Stat	11.400	11.400
Stock-Yogo 5% Critical Values	21.01	21.01
Stock-Yogo 10% Critical Values	11.52	11.52
Observations	606,213	606,213
Origin FE	Yes	Yes
Destination FE	Yes	Yes
Origin-Region FE	Yes	Yes
Distance Polynomials	Yes	Yes
Principal Components	Yes	Yes

Standard errors clustered at the country level.

	(5)	(6)	(7)
	Non-Gen Recent	Non-Gen Recent	Most Recent
$I_{o,-r(d)}^{1880} \times \frac{I_{-c(o),d}^{1880}}{I_{-c(o)}^{1880}} \times n_{r',o}$	0.000 (0.001)	0.000 (0.000)	-0.001*** (0.000)
$I_{o,-r(d)}^{1900} \times \frac{I_{-c(o),d}^{1900}}{I_{-c(o)}^{1900}} \times n_{r',o}$	0.002 (0.002)	0.002** (0.001)	0.001*** (0.000)
$I_{o,-r(d)}^{1910} \times \frac{I_{-c(o),d}^{1910}}{I_{-c(o)}^{1910}} \times n_{r',o}$	0.005 (0.006)	-0.001 (0.005)	-0.003* (0.001)
$I_{o,-r(d)}^{1920} \times \frac{I_{-c(o),d}^{1920}}{I_{-c(o)}^{1920}} \times n_{r',o}$	0.012 (0.010)	0.016 (0.014)	-0.001 (0.002)
$I_{o,-r(d)}^{1930} \times \frac{I_{-c(o),d}^{1930}}{I_{-c(o)}^{1930}} \times n_{r',o}$	-0.003 (0.005)	-0.006 (0.006)	0.002* (0.001)
$I_{o,-r(d)}^{1970} \times \frac{I_{-c(o),d}^{1970}}{I_{-c(o)}^{1970}} \times n_{r',o}$	0.001 (0.002)	-0.000 (0.002)	0.001 (0.000)
$I_{o,-r(d)}^{1980} \times \frac{I_{-c(o),d}^{1980}}{I_{-c(o)}^{1980}} \times n_{r',o}$	0.018** (0.007)	0.025*** (0.009)	-0.001 (0.002)
Kleibergen-Paap Wald RK F Stat	11.613	26.094	3.923
Stock-Yogo 5% Critical Values	21.01	21.01	21.01
Stock-Yogo 10% Critical Values	11.52	11.52	11.52
Observations	51,970,986	1,143,324	1,143,324
Technology FE	Yes	Yes	No
County FE	Yes	Yes	Yes
Country FE	No	No	Yes
Sample	All	Top 500 BT	Top 500 BT

Standard errors clustered at the county level.

Direction of Local Innovation: LPM

	1 Innovated $_{\tau,d}$		
	(1)	(2)	(3)
# Inputs Available $_{\tau',d}$	0.001*** (0.000)		
# US Inputs Available $_{\tau',d}$		0.000*** (0.000)	
# Foreign Inputs Available $_{\tau',d}$		0.002*** (0.001)	
1 Most Recently Invented Foreign Input Available $_{\tau',d}$			0.006*** (0.000)
Observations	51,970,986	51,970,986	51,970,986
Estimates	OLS	OLS	OLS
Technology FE	Yes	Yes	No
County FE	Yes	Yes	Yes
Country FE	No	No	Yes

Standard errors clustered at the county level.

Direction of Local Innovation: Shares

	11 Innovated $_{\tau,d}$	
	(1)	(2)
Share Inputs Available $_{\tau',d}$	5.019*** (0.103)	
Share US Inputs Available $_{\tau',d}$		3.768*** (0.118)
Share Foreign Inputs Available $_{\tau',d}$		2.527*** (0.072)
Observations	51,970,986	51,970,986
Estimates	Logit	Logit
Technology FE	No	No
County FE	No	No
Country FE	No	No
Sample	All	All

Standard errors clustered at the county level.

The Causal Effect of Local Idea Stocks on the Direction of Innovation

	1 Innovated $_{\tau,d}$		
	(1)	(2)	(3)
1 Most Recently Invented Foreign Input Available $_{\tau',d}$	0.006*** (0.000)	0.005*** (0.012)	0.033** (0.015)
Observations	51,970,986	51,970,986	342,369
Estimates	OLS	IV	IV
Technology FE	No	No	No
County FE	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Sample	All	All	$B \geq 25$

Standard errors clustered at the county level.

Technology Explanations (1/2)

- **CD-ROM (Japan):** The Compact Disc Read-Only Memory format was jointly developed by Sony (Japan) and Philips (Netherlands) in the early 1980s, building on their collaboration on audio CDs. Sony's contributions were central to the optical disc technology, laser reading systems, and error correction methods. Japanese electronics companies dominated CD-ROM drive manufacturing and the format became the standard for software distribution and data storage throughout the 1990s
- **Hybrid Car / Hybrid Electric Vehicle (Japan):** Hybrid electric vehicles combine a gasoline engine with an electric motor for improved fuel efficiency. The Toyota Prius, introduced in 1997, became the world's first mass-produced hybrid car and remains synonymous with hybrid technology. Japanese automakers (Toyota, Honda, Nissan) pioneered hybrid technology driven by Japan's energy efficiency priorities and automotive engineering excellence
- **Electron Microscope (Germany):** The electron microscope was invented by Ernst Ruska and Max Knoll at the Technical University of Berlin in 1931, earning Ruska the 1986 Nobel Prize in Physics. This breakthrough enabled visualization at near-atomic scales. German companies like Carl Zeiss and Leica remain global microscopy leaders. Germany's exceptional tradition in optics and precision instruments made the electron microscope possible
- **Polyvinyl Chloride (Germany):** Polyvinyl chloride (PVC) was first polymerized by German chemist Eugen Baumann in 1872, with industrial production pioneered by IG Farben in Germany during the 1920s-1930s. German chemical giants BASF, Hoechst, and Wacker developed commercial PVC manufacturing processes. PVC became one of the most widely used plastics globally, found in pipes, flooring, medical devices, and packaging. Germany's leadership in industrial chemistry made it the birthplace of modern plastics technology
- **Wireless LAN (Netherlands):** Wireless Local Area Network (WLAN) technology enables computers and devices to communicate over short distances without physical cabling. Foundational developments in wireless LAN were pioneered in the Netherlands in the early 1990s, particularly through research and engineering work at NCR in Delft. Dutch-led innovations laid the groundwork for the IEEE 802.11 standards, later branded as Wi-Fi, making the Netherlands a key origin country for modern wireless networking

Technology Explanations (2/2)

- **Antigen-Antibody Interaction (United Kingdom):** Specific antigen detection and immunoassay techniques were pioneered by British scientists. The UK's Medical Research Council and institutions like the National Institute for Medical Research made crucial advances in understanding antigen-antibody specificity. British researchers developed foundational methods for diagnostic testing, including contributions to radioimmunoassay and ELISA techniques that revolutionized clinical diagnostics.
- **Cloud Computing (Sweden):** Sweden made early contributions to distributed computing and internet infrastructure that laid groundwork for cloud computing. Swedish companies and research institutions pioneered data center technology and virtualization. Ericsson contributed to network infrastructure enabling cloud services. Sweden's strong IT sector and early internet adoption fostered innovations in scalable computing architectures.
- **Blu-ray (South Korea):** Blu-ray Disc is a high-definition optical disc format developed to succeed DVD. Samsung and LG, both Korean electronics giants, were founding members of the Blu-ray Disc Association alongside Sony and Panasonic. South Korea's dominance in consumer electronics, semiconductor manufacturing, and display technology made it a natural leader in next-generation optical media.
- **GPRS Roaming Exchange (Finland):** GPRS (General Packet Radio Service) roaming technology enables mobile data connectivity across different cellular networks globally. Finland, home to Nokia, was a pioneer in mobile telecommunications and GSM/GPRS infrastructure development. Nokia Networks (now Nokia Solutions and Networks) was instrumental in developing and deploying GPRS roaming exchanges that allowed seamless mobile data services across international borders. Finland's telecommunications expertise and Nokia's dominance in mobile network equipment made it a leader in GPRS standardization and deployment in the late 1990s and early 2000s.

Identifying Assumption

Any confounding factors that make a given destination more attractive for both migration and innovation from a given origin country do not simultaneously affect the interaction of the settlement of migrants from other continents with the total number of migrants arriving from the same origin but settling in a different census region.

- Usually also include $\delta_d \times \delta_{c(o)}$ and $\delta_o \times \delta_{r(d)}$ fixed effects.
- Remaining challenge: a common unobserved characteristic of destinations in two different census regions, correlated with innovation, may still have disproportionately caused large groups of migrants from two origins on two different continents to *simultaneously* migrate to the same destinations across census regions.

Robustness: Alternative Specifications

	Log # Foreign Technologies $_{o,d}$				# Foreign Techs $_{o,d}$
	(1)	(2)	(3)	(4)	(5)
Log Ancestry $_{o,d}$	0.379*** (0.131)	0.464*** (0.150)	0.333*** (0.128)	1.062** (0.480)	
Ancestry $_{o,d}$					0.429** (0.187)
Observations	606,213	606,213	606,213	53,397	606,213
Estimates	IV	IV	IV	IV	IV
Origin FE	Yes	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes	Yes
Origin-Region FE	Yes	Yes	Yes	Yes	Yes
Distance Polynomials	Yes	Yes	Yes	Yes	Yes
Robustness	Baseline	5yr Threshold	Post-1980	Innovating	Levels

Standard errors clustered at the country level.

Robustness: Excluding Top Origins and Destinations

	Log # Foreign Technologies $_{o,d}$				
	(1)	(2)	(3)	(4)	(5)
Log Ancestry $_{o,d}$	0.379*** (0.131)	0.383*** (0.133)	0.378*** (0.131)	0.352*** (0.131)	0.310** (0.123)
Observations	606,213	595,019	604,862	603,072	603,072
Estimates	IV	IV	IV	IV	IV
Origin FE	Yes	Yes	Yes	Yes	Yes
Destination FE	Yes	Yes	Yes	Yes	Yes
Origin-Region FE	Yes	Yes	Yes	Yes	Yes
Distance Polynomials	Yes	Yes	Yes	Yes	Yes
Robustness	Baseline	No NE Corridor	No Silicon Valley	No Japan	No Germany

Standard errors clustered at the country level.

Direction of Local Innovation

	1 Innovated $_{\tau,d}$		
	(1)	(2)	(3)
# Inputs Available $_{\tau',d}$	0.031*** (0.001)		
# US Inputs Available $_{\tau',d}$		0.013*** (0.001)	
# Foreign Inputs Available $_{\tau',d}$		0.152*** (0.006)	
1 Foreign Input Available (Most Recent) $_{\tau',d}$			3.510*** (0.090)
Baseline $P(\text{adopt})$	0.34%	0.28%	0.48%
Marginal Effect	0.013p.p.	0.066p.p.	5.00p.p.
Sample	All	All	All
Observations	19,967,337	19,967,337	19,967,337
Estimates	Logit	Logit	Logit

Standard errors clustered at the county level.

The Causal Effect of Local Idea Stocks on the Direction of Innovation

	1 Innovated $_{\tau,d}$			
	Logit	Logit CF		LPM IV
	(1)	(2)	(3)	(4)
1 Foreign Input Available (Most Recent) $_{\tau',d}$	3.510*** (0.090)	7.124*** (0.125)	6.766*** (0.185)	0.536*** (0.163)
Control Function Residual		-6.637*** (0.164)	-5.725*** (0.275)	
Baseline $P(\text{adopt})$	0.48%	0.48%	0.50%	0.50%
Marginal Effect	5.00p.p.	50.94p.p.	38.78p.p.	53.56p.p.
County FE	No	No	No	Yes
Country FE	No	No	No	Yes
Sample	All	All	$B \geq 25$	$B \geq 25$
Observations	19,967,337	19,967,337	106,794	106,794

Standard errors clustered at the county level.